

PROJECT TITAN

Requirements Traceability Matrix

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EDUCATIONAL USE NOTICE

Project TITAN is an educational engineering design project. It is not a clinical device, regulatory submission, or product approved for human use.

Document Control

Revision	Date	Description	Author
A	July 2026	Initial traceability release linking requirements to current evidence and open verification work.	Michael Lozada-Longog

1. Traceability Matrix

Req ID	Requirement	Method	Evidence	Status
FR-001	Two-bearing shaft support	CAD inspection	Assembly model; 31 mm bearing-center spacing evidence	Verified
FR-002	Interference-free modular assembly	CAD review	Final assembled model and exploded view	Verified
FR-003	Replaceable components	Architecture review	Separate component instances and drawings	Verified by design
PR-001	Minimum SF ≥ 2.0	Static FEA	Reported minimum SF 2.377	Verified for current study
PR-002	Stress below configured yield	Static FEA + material audit	54.947 MPa max; Fusion effective yield implied by SF is ~ 130.6 MPa	Conditional - audit required
PR-003	Displacement ≤ 0.005 in	Static FEA	0.004 in maximum	Verified
MR-001	CNC-compatible parts	Drawing/process review	Milling and turning plan	Verified by review
MR-002	Commercial bearings/hardware	BOM review	Bearing geometry present; exact catalog selections open	Partially verified
DR-001	Part drawings	Document review	Completed individual drawing files	Verified
DR-002	Engineering report	Document review	Project TITAN Engineering Design Report.pdf	Verified
VR-001	Record loads and geometry	Evidence review	2 x 2000 N screenshots; 6 mm thickness confirmation	Verified
VR-002	Analytical sanity checks	Calculation review	CAL-001	Verified with limitations
VR-003	Physical validation	Prototype test	Future test report	Open / future

2. Closure Summary

The current V1 package closes CAD, documentation, and static-FEA requirements. Material-model confirmation and physical validation remain open. Those open items are disclosed rather than treated as completed evidence.